

MAKUSHINO, Russian Federation

WMO: 286660

Lat: 55.250N

Lon: 67.300E

Elev: 140

StdP: 99.65

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.1	-29.6	-36.3	0.1	-32.9	-32.9	0.2	-29.6	11.5	-5.6	10.0	-10.5	1.8	200	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	31.3	19.3	29.5	19.1	27.7	18.4	21.0	28.0	20.2	27.1	19.4	25.9	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	23.3	17.9	13.0	22.7	17.0	12.3	22.4	61.3	27.8	58.5	27.2	56.0	25.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.4	8.2	-35.4	34.2	3.2	1.8	-37.7	35.5	-39.6	36.6	-41.4	37.6	-43.7	38.9		
(5)			-35.1	23.1	3.2	1.4	-37.4	24.1	-39.2	24.9	-41.0	25.7	-43.3	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-17.3	-14.8	-6.6	4.6	13.0	18.1	19.7	17.3	11.3	3.7	-5.9	-13.4
	DBStd	14.26	7.07	6.62	5.31	5.66	5.26	4.72	3.88	4.50	4.59	5.03	7.27	7.59
	HDD10.0	3708	845	695	513	178	30	3	0	2	38	201	477	725
	HDD18.3	5936	1103	928	772	411	177	61	32	75	214	452	727	983
	CDD10.0	1002	0	0	0	17	124	246	302	228	78	7	0	0
	CDD18.3	188	0	0	0	0	12	54	76	43	4	0	0	0
	CDH23.3	1942	0	0	0	5	164	569	743	418	42	0	0	0
	CDH26.7	613	0	0	0	1	44	192	235	135	6	0	0	0
Wind	WSAvg	3.8	3.7	3.8	4.1	4.3	4.0	3.7	3.1	3.3	3.7	3.9	4.0	4.0
Precipitation	PrecAvg	366	16	11	13	19	31	56	68	47	33	28	24	19
	PrecMax	552	33	29	45	48	76	119	160	99	104	100	44	66
	PrecMin	178	1	1	1	1	3	6	12	4	0	2	0	0
	PrecStd	81	7	7	10	13	19	33	38	23	22	22	11	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	1.5	8.2	23.4	30.6	33.0	33.5	32.6	27.7	19.3	8.3	1.8
		MCWB	-0.7	-0.1	5.0	13.2	17.4	19.4	20.2	20.1	15.3	11.3	5.8	0.3
	2%	DB	-2.2	-0.5	3.8	19.2	27.4	31.0	31.1	30.2	24.3	16.1	5.8	0.1
		MCWB	-2.9	-1.5	1.7	10.8	16.4	18.9	19.9	19.4	14.3	9.4	4.2	-0.7
	5%	DB	-4.5	-2.6	2.2	16.3	24.9	29.1	29.2	28.0	21.4	13.5	4.1	-1.6
		MCWB	-5.1	-3.4	0.7	9.8	15.4	19.0	19.5	18.7	13.9	8.4	2.6	-2.4
	10%	DB	-6.7	-5.0	1.0	13.5	22.3	26.8	27.4	25.6	18.7	11.0	2.0	-3.4
		MCWB	-7.3	-5.8	-0.3	8.7	13.8	18.1	19.0	17.5	12.6	7.2	0.7	-4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	0.2	5.0	13.9	19.8	21.7	22.4	21.2	17.1	12.1	6.2	0.5
		MCDB	-0.2	1.1	7.9	22.0	27.8	28.8	28.3	29.1	24.2	17.8	7.4	1.2
	2%	WB	-2.9	-1.5	2.1	11.8	17.4	20.4	21.3	20.2	15.5	10.5	4.5	-0.6
		MCDB	-2.2	-0.5	3.5	16.7	25.4	28.0	27.7	27.7	21.6	14.9	5.8	0.2
	5%	WB	-5.1	-3.4	0.8	10.4	15.9	19.4	20.5	19.5	14.3	8.9	2.6	-2.3
		MCDB	-4.5	-2.6	2.0	16.1	23.1	26.8	27.0	25.8	19.9	12.6	4.0	-1.6
	10%	WB	-7.3	-5.8	-0.2	8.7	14.5	18.5	19.8	18.4	13.1	7.5	0.9	-4.2
		MCDB	-6.7	-5.0	1.0	13.4	21.2	25.6	26.0	24.2	18.2	10.7	1.7	-3.5
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.3	8.4	8.9	11.3	10.8	10.8	10.4	9.4	7.4	6.0	7.0
		MCDBR	7.6	8.3	7.5	13.3	14.6	13.9	13.4	13.5	13.3	11.8	6.1	6.9
		MCWBR	7.3	7.8	6.0	6.9	6.5	5.6	4.9	5.1	6.1	6.8	5.1	6.5
	5% WB	MCDBR	7.5	8.4	7.1	12.4	13.0	12.6	11.7	12.3	11.8	10.6	5.8	6.9
		MCWBR	7.3	7.9	5.9	6.9	6.6	5.7	5.1	5.2	6.1	6.6	5.1	6.6
Clear-Sky Solar Irradiance	taub	0.282	0.317	0.370	0.411	0.407	0.404	0.424	0.433	0.383	0.357	0.318	0.284	
	taud	2.054	2.020	1.957	2.112	2.231	2.313	2.255	2.195	2.292	2.232	2.121	2.019	
	Ebn at Noon	608	725	770	804	835	842	814	776	768	682	557	498	
	Edh at Noon	67	101	137	136	128	120	125	125	99	82	63	55	
All-Sky Solar Radiation	RadAvg	0.94	2.00	3.56	4.55	5.53	5.96	5.60	4.47	3.05	1.54	0.75	0.59	
	RadStd	0.07	0.09	0.23	0.48	0.44	0.55	0.47	0.45	0.24	0.25	0.10	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page